JIANXUAN SHAO

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RESEARCH INTERESTS

Human-AI Collaboration | Context-Aware Personal AI | Multimodal and Spatial Interaction

I aim to study personal AI that understands ongoing activities, converses and reasons with people, connects decisions to useful actions, and adapts through repeated collaboration. My preparation combines architectural design, immersive installations, interactive prototypes and reproducible computational evaluation.

EDUCATION

University of Sydney | Feb 2025 - Present

Master of Computer Science | *Focus: Data Science and AI* | *Expected Dec 2026*

Average across 12 graded units: 79.33/100; Semester 1, 2026 average: 84.5/100. Selected results: Machine Learning and Data Mining 89 HD; Introduction to Programming 87 HD; Computer and Network Organisation 85 HD; Introduction to Cybersecurity 92 HD.

Hebei University | Sep 2018 - Jun 2023

Bachelor of Engineering in Architecture | *GPA 4.05/5.0* | *Average 86.0/100*

Five-year programme covering architectural design, building physics, computational design and digital visualisation.

RESEARCH AND INTERACTIVE SYSTEMS

Game Fellows | 2025 - Present

Prototype development and study preparation | *Python / Pygame*

- Built following and fetching prototypes, then extended a Galaga-style game with a visible teammate for formation, projectile interception, protection and coordinated support fire.
- Implemented bounded movement, stable behaviour states, cooldowns and action-linked text/audio cues. The reproducible build uses local game-state rules; screenshot-to-AI interpretation was explored separately.
- Prepared counterbalanced Solo/Fellow trials with paired random seeds, gameplay/event logs and questionnaires on presence, trust, usefulness and player control. Human-participant evaluation has not yet been conducted.

Explainable AI-Generated Image Detection | 2026 - Present

Research design, baseline verification and error analysis | *Eight-person capstone team*

- Contributed to research-question development and experimental planning, and worked on the direct-detection baseline and reproducibility checks. Teammates implemented or evaluated other modules and integrated the fusion system.
- Audited 60,000 predictions from 10 model runs against a canonical manifest. Reconstructed all 900 fused decisions from frozen outputs and coefficients, categorised 51 errors, and documented evaluation limitations.
- The team system combined Qwen3.5-9B reasoning and ResNet-50 visual forensics. Model selection had included the internal evaluation split; external testing and human evaluation of explanations remain future work.

SPATIAL SYSTEMS AND INTERACTION EXPERIENCE

Shanghai Fengyuzhu Culture Technology Co., Ltd. | Jul 2023 - Jul 2024

Interactive Media Intern

- Worked on the 2023 Damo Cave immersive exhibition through spatial layout, lighting placement and tuning, projection content, equipment configuration and on-site integration.
- Participated in setting up and debugging multi-channel projection blending, wall-touch control and radar recognition, checking display and interaction components together on site.
- Applied VR/AR visualisation and digital storytelling to architectural concepts in the Dragon Roaming by the Riverside cultural tourism project.

Browser Interaction Prototypes | 2026

AI-assisted development | JavaScript / Canvas / Three.js / MediaPipe

- Developed a Damo Cave browser exploration linking hand/body input with projection brightness, scene reveals and movement-driven floor effects, with mouse fallback. This later experiment is separate from the 2023 installation.
- Built a browser-based 3D demonstration of action previews, confirmation, undo, pinned objects and scoped rules. Headset testing has not been conducted.

Daci Pavilion and Geyuan Temple | Oct 2021

Survey and reconstruction team | Hebei University

- Recorded historic buildings through field sketches and CAD/Rhino documentation; reconstructed plans, elevations, sections and 3D geometry, including timber bracket components and architectural details.

SELECTED COMPUTATIONAL PROJECTS

Classification and PathMNIST Pipelines | 2026

Built preprocessing, stratified cross-validation and tuning pipelines; compared Random Forest, MLP and CNN models using macro-F1, class-level errors and runtime. Machine Learning and Data Mining coursework.

Electricity Data Engineering and Spatial Integration | 2026

Integrated geocoded energy and regional datasets in spatial DuckDB; processed 716,454 facility records and streamed JSON via MQTT to a Streamlit map dashboard. Data Engineering coursework.

TECHNICAL AND RESEARCH SKILLS

Programming and data: Python, SQL, Pygame, DuckDB, Streamlit, MQTT; preprocessing, model evaluation and reproducible analysis.

Interaction and spatial design: State-based behaviour, text/audio feedback, event logging, CAD, Rhino/Grasshopper, 3D modelling, projection and lighting integration.

Browser prototyping: AI-assisted JavaScript, Canvas, Three.js and MediaPipe development.

Research practice: Research-question development, controlled study preparation, comparative evaluation, error analysis and technical reporting.

ADDITIONAL EXPERIENCE AND AWARDS

Hebei Haoyu, Beidaihe | Architectural Design Intern | Jun - Aug 2022: school layout studies, design iterations and drawings. Qinhuangdao Guangjian | Assistant Architect Intern | Jun - Aug 2021: redevelopment drawings.

Hebei University Scholarship: Second Prize (2019), Third Prize (2021, 2022). JYZC Innovation & Creativity Contest: Nomination (2022). Tennis Club President (2019 - 2020); campus architectural sketching team leader (2022).